MUHAMMAD ZAIN-UL-ABIDEEN

AI Engineer

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Summary

AI Engineer with two internships and four deployed AI products in speech, computer vision, RAG and LLM agents. I build the whole system, from data and model to backend and frontend, and test it on real-world data.

Education

National University of Sciences and Technology (NUST) Sep 2022 – Jun 2026
Bachelor of Electrical Engineering Islamabad, Pakistan

Technical Skills

Languages & Fundamentals: Python, TypeScript, JavaScript, SQL, Data Structures & Algorithms
Machine Learning: Deep Learning, PyTorch, Scikit-learn, Hugging Face Transformers, Pandas, NumPy, Computer Vision, NLP, Speech Recognition (Whisper), CLIP, ONNX Runtime, Model Evaluation
LLMs & Retrieval: LLM Applications, RAG, AI Agents, Prompt Engineering, Embeddings, Vector Search (pgvector, ChromaDB, FAISS), Gemini, Groq, Langfuse
Backend & Cloud: FastAPI, REST APIs, PostgreSQL, Next.js, Docker, GitHub Actions (CI/CD), pytest, Git, Microsoft Azure, Google Cloud, Vercel, Render

Projects

Earshot – Podcast Search with Audio Citations | Python, Whisper, PostgreSQL earshot-ten-pi.vercel.app

- Built a search engine for podcasts: you ask a question, get a short answer, and every quote links to the exact second it was said, so one click plays the original audio.
- Compared keyword, semantic and reranked search on a test set I wrote; a reranker raised top-5 accuracy from 76% to 93%, but it was too slow for the free server, so I shipped the faster version and documented the trade-off.
- Wrote a checker that matches every quote word for word against the transcript; when the model cannot back up an answer, the app says so instead of guessing.

Dehqan – Crop Disease Doctor for Pakistani Farmers | Python, PyTorch, CLIP, ONNX dehqan.vercel.app

- Trained a leaf-disease model for 13 crops and tested it on real farm photos, not just lab images; it fell to 12% accuracy in the field, so I rebuilt it on CLIP with field data and raised it to 62%.
- Taught it to say "not sure" and refer farmers to an extension office when confidence is low, added a check that rejects photos that are not leaves, and shrank the models with ONNX to run on a free server.
- Added Urdu and English voice chat, safe spray times from a live weather forecast, and a guardrail that refuses every pesticide-dose question rather than risk wrong advice.

Rehnuma – AI Assistant for Electricity Bills | Python, FastAPI, Gemini, Next.js rehnuma-kappa.vercel.app

- Built an Urdu/English assistant that reads a photo of a Pakistani electricity bill and re-checks every charge with its own calculator, so the AI never does the maths; it reads 96% of fields correctly on real bills.
- Answers questions about NEPRA regulations with the exact clause cited, and rewrites Urdu questions into the regulations' wording so Urdu search works; tested with solar and non-solar households, with evaluation checks run on every code change.

Parity – Web Accessibility Auditor | Python, FastAPI, Playwright, Gemini parity-iota-puce.vercel.app

- Built a tool that opens a website in a real browser and checks whether people with disabilities can use it, pairing standard WCAG rule checks with a vision model that judges whether image descriptions make sense.
- Noticed the vision model was flagging correct descriptions as wrong and added a consistency check that raised its precision from 80% to 100%; it runs as a queued web service with rate limits and protection against malicious URLs.

Experience

Machine Learning Engineer Intern Jun 2025 – Aug 2025
RISETech Islamabad, Pakistan

- Collected sensor data outdoors and designed the features for a vehicle classifier that uses vibration, magnetic and acoustic signals; ensemble models built on those features raised accuracy from 71% to 99%.
- Wrote up the feature extraction method as an IEEE-format research paper.

AI Engineer Intern Jun 2024 – Aug 2024
Software Productivity Strategists (SPS), NSTP Islamabad, Pakistan

- Built Python backend features for two company products (BMS and CSM) and deployed AI features on Microsoft Azure.

Certifications

Google AI Professional Certificate | Advanced ML on Google Cloud Specialization | IBM Machine Learning Professional Certificate | AWS Cloud Practitioner Essentials